

THEORY AND METHODS

A BAYESIAN NON-PARAMETRIC APPROACH TO THE MIXED LINEAR MODEL WITH AN APPLICATION TO ANIMAL BREEDING

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Summary: In this article, the Dirichlet process prior is used to provide non-parametric Bayesian estimates for the random effects in the case of the mixed linear model. Since the random effects are typically assumed to be normally distributed in both the Bayesian and classical models, we provide a Gibbs sampler algorithm that allows these effects to have a non-parametric prior distribution. The full conditional distributions are given as well as the complete Gibbs sampler. To illustrate the application, data from Duchateau et al. (1998), a total of 1277 weaning weights from the progeny of 200 sires are used.

1. Introduction

In animal breeding applications, it is usually assumed that the data follows a mixed linear model. Mixed linear models are naturally modelled within the Bayesian framework. The main advantage of a Bayesian approach is that it allows explicit use of prior information, thereby giving new insights in problems where classical statistics fail. Harville (1990) stated “A more extensive use of Bayesian ideas by animal breeders and other practitioners is

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desirable and is more feasible from a computational standpoint than commonly thought. The Bayesian approach can be used to devise prediction procedures that are more sensible – from both a Bayesian and Frequentist perspective – than those in current use.”

As mentioned by Duchateau, Janssen and Rowlands (1998) “The emphasis in breeding experiments is on the variance components and on the prediction of particular random effects, but estimation of fixed effects and hypothesis testing with regard to fixed effects is also important.”

An animal breeding experiment is used to illustrate this. A part of the data set can be found in Example 3.2 of Duchateau et al. (1998). The goal of the research was to select for improved helminth resistance in sheep. The female sheep used in the experiment were from three different breeds, whereas the males were from two breeds. In each of the six crosses, there were at least 25 and at most 42 different sires, and each sire within a cross breed had an average offspring of 6.4 lambs. The weaning weight was measured for each of the lambs. The age at which lambs were weaned may differ from animal to animal, and therefore, a variable expressing the age of the animal at weaning was included as a fixed effect in the model. Also the sex of the lamb was added to the model as a fixed effect. Finally, “sire” was included as a random effect u_i . Duchateau et al. (1988) use “sire” to refer to a male-female combination. It was further assumed that these random effects are independent.

In what follows vectors and matrices will be denoted by lower and uppercase bold symbols respectively. An appropriate mixed linear model for this data structure is thus given by

$$y_i = X_i\beta + z_iu_i + \varepsilon_i \quad (i = 1, \dots, q) \quad (1.1)$$

where y_i is $n_i \times 1$, the vector of weaning weights for the lambs (progeny) of the i th sire, X_i : known incidence matrix of order $n_i \times p$, z_i : vector of n_i elements 1, β : $p \times 1$ vector of uniquely defined fixed effects (so that X_i has full column rank p), u_i the random effect of the i th sire and ε_i : $n_i \times 1$ vector of random residuals. It is standard in implementations of this model to assume ε_i and u_i are independent and that both are normally distributed with $\varepsilon_i \sim N_{n_i}(0, \sigma_\varepsilon^2 I_{n_i})$ and $u_i \sim N(0, \sigma_u^2)$, where I_s is the $s \times s$ identity matrix and $N_s(\mu, \Sigma)$ denotes the s -dimensional multivariate normal distribution with mean μ and covariance matrix Σ . Under these assumptions

$$y_i | \beta, u_i, \sigma_\varepsilon^2 \sim N_{n_i}(X_i \beta + z_i u_i, \sigma_\varepsilon^2 I_{n_i}). \quad (1.2)$$

As mentioned above, the distribution of the random effects is usually taken to be normal. Large-sample classical (frequentist) inference regarding β should not be much affected asymptotically by changing the distribution of the random effects since the first two moments of the marginal distribution of y_i do not depend on the normality of the distribution of the random effects. To see this, notice that, marginally

$$y_i | \beta, \sigma_\varepsilon^2, \sigma_u^2 \sim N_{n_i}(X_i \beta, (z_i z_i') \sigma_u^2 + \sigma_\varepsilon^2 I_{n_i}). \quad (1.3)$$

For inference regarding β from (1.3), the individual regression coefficients, simply result in a complex covariance structure. Bayesian inference for β , using the marginal likelihood implied by (1.3) will depend only on $(y, \sigma_\varepsilon^2, \sigma_u^2)$ but the nature of this dependence will be sensitive to the distributional form ascribed to the u_i .

As in Kleinman and Ibrahim (1998) we will present a model in this note for which the random effects have a non-parametric distribution. The non-parametric Bayesian approach for the random effects u_i is to specify a prior distribution on the space of all possible distribution functions. This prior is

applied to the general prior distribution for the u_i . This can be accomplished with a Dirichlet process prior distribution. For the mixed linear model, this means that the usual normal prior on the random effects is replaced with a non-parametric prior, followed by a Dirichlet process prior on the general distribution. The foundation of this technology is discussed in Ferguson (1973), where the Dirichlet process and its usefulness as a prior distribution are discussed. The practical application of such models, using the Gibbs sampler, has been pioneered by Doss (1994), MacEachern (1994), Escobar (1994), Bush and MacEachern (1996), Liu (1966) and Müller, Erkanı and West (1996). Other important work in this area has been done by West et al. (1994), Escobar and West (1995) and MacEachern and Müller (1998).

Escobar (1991 and 1994) showed that for the random effects model a prior based on a finite mixture of Dirichlet processes leads to an estimator that has excellent behaviour. He compared this estimator to standard estimators under two distinct priors. When the prior of the random effects is normal, his estimator performs nearly as well as the standard Bayes estimator that requires the estimate of the prior to be normal. When the prior is a two-point distribution, his estimator performs nearly as well as a non-parametric maximum likelihood estimator.

Mixture of Dirichlet process priors can be of great importance in animal breeding experiments and will hopefully contribute in solving the undeclared preferential treatment of animals problem. Prof. Daniel Gianola, well-known animal breeder from the University of Wisconsin, USA, mentioned that "A serious problem in animal breeding is the undeclared preferential treatment of animals which have an unusual economic value. As long as genetic evaluation systems lack information about such preferential treatment, predictions of breeding values of favoured animals obtained with mixed

Gaussian models are inflated, whereas those of other animals are deflated. So far, this problem has not been solved satisfactorily.” Prof. Gianola suggested a “robust” mixed effects linear model based on the t -distribution for the “preferential treatment problem”. The Gibbs sampler will then employ a data augmentation strategy where auxiliary Gaussian distributions with randomly varying dispersion parameters are introduced, thus providing an analysis that is robust to both data outliers and outlying individuals in the population.

The t -distribution, however, does not cover departures from symmetry, while the Dirichlet process prior will be able to do so. Despite the attractive features of the Dirichlet process, it has only recently been investigated. The reason for this delay is, as has been mentioned, the computational difficulties involved with mixtures of Dirichlet processes. Only recently have tools allowing Bayesian analysis become computationally feasible. For the mixed linear model the unconditional posterior distributions can be obtained by using Gibbs sampling and data augmentation.

The rest of this manuscript is organised as follows. In Section 2 we show how to apply the mixture of Dirichlet process (MDP) structure, in Section 3 we discuss the simulation procedure for the precision parameter and in Section 4 we illustrate the methodology with real data. We must, however, state that the reference to Gianola has no linkage to this particular data set problem. The data set of Duchateau et al. (1998) is included mainly for illustration and comparison purposes; i.e. to compare the corresponding results of the REML, Traditional Bayes and Non-parametric Bayesian procedures. We are, however, quite sure that the Dirichlet process prior will play an important role in cases where undeclared preferential treatment of animals occur.

2. Dirichlet process priors in the case of the mixed linear model

In model (1.2) and for our data set, “flat” or uniform prior distributions are assigned to σ_ϵ^2 and β as to represent lack of prior knowledge about the vector of fixed effects β and the error variance σ_ϵ^2 .

Therefore

$$p(\beta, \sigma_\epsilon^2) = p(\beta) p(\sigma_\epsilon^2) \propto \text{constant}. \quad (2.1)$$

Further, the prior distribution for the random effects u_i ($i = 1, \dots, q$) is given by

$$u_i \sim G. \quad (2.2)$$

and independently of each other, where

$$G \sim DP(M \cdot G_0). \quad (2.3)$$

Such a model assumes that the prior distribution G itself is uncertain, drawn from a Dirichlet process. The parameters of a Dirichlet process are G_0 , a probability measure, and M , a positive scalar assigning mass to the real line. The parameter G_0 , often called the base measure or base prior, is a distribution that approximates the true nonparametric shape of G ; it can therefore be assumed that $E(G) = G_0$ (see West et al., 1994). The parameter M on the other hand reflects our prior belief about how similar the nonparametric distribution G is to the base measure G_0 . There are two special cases in which the mixture of Dirichlet process (MDP) models leads to the fully parametric case. As $M \rightarrow \infty$, $G \rightarrow G_0$ so that the base prior is the prior distribution for u_i . Also if $u_i \equiv u$ for all i , the same is true. In this note we will assume that

$$G_0 = N(0, \sigma_u^2). \quad (2.4)$$

In many Bayesian problems, marginal posterior distributions are often needed to make appropriate inferences. However, due to the complexity of the joint posterior distribution it is impossible to obtain these marginal densities analytically and because of the many unknowns very difficult to calculate numerically.

Recently, due to work of Gelfand and Smith (1990), Gelfand et al. (1990), Carlin et al. (1992) and Gelfand et al. (1992), the Gibbs sampler has been shown as a useful tool for applied Bayesian inference in a broad variety of statistical problems. The Gibbs sampler is implicit in the work of Hastings (1970) and made popular in the image processing context by Geman and Geman (1984). the Gibbs sampler is an adaptive Monte Carlo integration technique. The typical objective of the sampler is to collect a sufficiently large enough number of parameter realizations from conditional posterior densities in order to obtain accurate estimates of the marginal posterior densities. The principal requirement of the sampler is that all conditional densities must be available in the sense that random variates can be generated from them. The method is of great appeal on account of its simple logical foundation and reasonable ease of implementation.

When G is a fully parametric prior, the joint posterior can easily be found as in Wilks et al. (1993). The fully conditional posterior density for each of the unknowns is obtained by regarding all other parameters in the joint posterior as known. The required conditionals for β and σ_ϵ^2 are

$$\beta | u, \sigma_\epsilon^2, y \sim N_p \left\{ \hat{\beta}, \sigma_\epsilon^2 \left(\sum_{i=1}^q (X_i' X_i) \right)^{-1} \right\} \quad (2.5)$$

where

$$\hat{\beta} = \left(\sum_{i=1}^q (X_i' X_i) \right)^{-1} \sum_{i=1}^q X_i' (y_i - z_i u_i)$$

and

$$p(\sigma_\varepsilon^2 | \beta, \mathbf{u}, \mathbf{y}) = K_\varepsilon \left\{ \prod_{i=1}^q \left(\frac{1}{\sigma_\varepsilon^2} \right)^{n_i/2} \right\} \exp \left\{ -\frac{1}{2\sigma_\varepsilon^2} \sum_{i=1}^q (\mathbf{y}_i - \mathbf{X}_i\beta - \mathbf{z}_i\mathbf{u})' (\mathbf{y}_i - \mathbf{X}_i\beta - \mathbf{z}_i\mathbf{u}) \right\}$$

$$\sigma_\varepsilon^2 > 0 \quad (2.6)$$

an inverse gamma density, where

$$K_\varepsilon = \left\{ \frac{\sum_{i=1}^q (\mathbf{y}_i - \mathbf{X}_i\beta - \mathbf{z}_i\mathbf{u}_i)' (\mathbf{y}_i - \mathbf{X}_i\beta - \mathbf{z}_i\mathbf{u}_i)}{2} \right\}^{\frac{1}{2}(n-2)} \frac{1}{\Gamma\left(\frac{n-2}{2}\right)}.$$

Also $\mathbf{u} = (u_1 \ u_2 \ \dots \ u_q)'$, $\mathbf{y} = (y_1' \ y_2' \ \dots \ y_q')'$ and $n = \sum_{i=1}^q n_i$, the sample size.

From the discussion of MDP models in Section 2 of Kleinman and Ibrahim (1998), we find, as in West et al. (1994), see also Escobar (1994) that the conditional posterior of u_i is given by

$$p(u_i | \beta, \sigma_\varepsilon^2, \sigma_u^2, \mathbf{u}_{(i)}, M) \propto \sum_{j \neq i}^q \phi(y_i | \mathbf{X}_i\beta + \mathbf{z}_i u_j, \sigma_\varepsilon^2 \mathbf{I}_{n_i}) \cdot \delta_{u_j}$$

$$+ \left\{ M \int_{-\infty}^{\infty} \phi(y_i | \mathbf{X}_i\beta + \mathbf{z}_i u_i, \sigma_\varepsilon^2 \mathbf{I}_{n_i}) \phi(u_i | 0, \sigma_u^2) du_i \right\}$$

$$\times \phi(u_i | 0, \sigma_u^2) p(y_i | u_i, \beta, \sigma_\varepsilon^2, y_j) \quad (2.7)$$

where $p(y_i | u_i, \beta, \sigma_\varepsilon^2, y_j) = \phi(y_i | \mathbf{X}_i\beta + \mathbf{z}_i u_i, \sigma_\varepsilon^2 \mathbf{I}_{n_i})$ and $\phi(\cdot | \mu, \sigma^2)$ denotes the normal density with mean μ and variance σ^2 . Also $\mathbf{u}_{(i)}$ denotes the vector of random effects for the subjects (sires) excluding subject i and δ_s is a degenerate distribution with point mass at s .

Consider the integral

$$\begin{aligned}
 I_i &= M \int_{-\infty}^{\infty} \phi(y_i | X_i \beta + z_i u_i, \sigma_\varepsilon^2 I_{n_i}) \phi(u_i | 0, \sigma_u^2) du_i \\
 &= M \int_{-\infty}^{\infty} \left(\frac{1}{2\sigma_\varepsilon^2 \pi} \right)^{\frac{1}{2} n_i} \exp \left\{ -\frac{1}{2\sigma_\varepsilon^2} (y_i - X_i \beta - z_i u_i)' (y_i - X_i \beta - z_i u_i) \right\} \times \\
 &\quad \left(\frac{1}{2\sigma_u^2 \pi} \right)^{\frac{1}{2}} \exp \left\{ -\frac{1}{2} u_i^2 / \sigma_u^2 \right\} du_i.
 \end{aligned}$$

After completing the square with respect to u_i , it follows that

$$I_i = M (2\pi)^{-\frac{1}{2} n_i} (\sigma_u^2)^{-\frac{1}{2}} Q_i^{\frac{1}{2}} (\sigma_\varepsilon^2)^{-\frac{1}{2} n_i} \exp \left\{ \frac{1}{2\sigma_\varepsilon^2} (y_i - X_i \beta)' U_i (y_i - X_i \beta) \right\}$$

where

$$Q_i = \left[\frac{1}{\sigma_\varepsilon^2} (z_i' z_i) + \frac{1}{\sigma_u^2} \right]^{-1} \quad \text{and} \quad U_i = \left(\frac{1}{\sigma_\varepsilon^2} z_i Q_i z_i' - I_{n_i} \right). \quad (2.8)$$

Each summand in the conditional posterior distribution of u_i is separated into two elements. The first element is a mixing probability, and the second is a distribution to be mixed. Simulation of u_i from its conditional posterior distribution can therefore be obtained in the following way:

$p \{ u_i | \beta, \sigma_\varepsilon^2, \sigma_u^2, \mathbf{u}_{(i)}, y, M \} = u_j$ with probability

$$\begin{aligned}
 &\frac{\phi(y_i | X_i \beta + z_i u_j; \sigma_\varepsilon^2 I_{n_i})}{I_i + \sum_{\substack{j=1 \\ j \neq i}}^q \phi(y_i | X_i \beta + z_i u_j; \sigma_\varepsilon^2 I_{n_i})} \\
 &\sim h(u_i | \beta, \sigma_\varepsilon^2, \sigma_u^2, \mathbf{u}_{(i)}, y_i) \quad \text{with probability}
 \end{aligned}$$

$$\frac{I_i}{I_i + \sum_{\substack{j=1 \\ j \neq i}}^q \phi(y_i | X_i \beta + z_i u_j; \sigma_\epsilon^2 I_{n_i})} \quad (2.9)$$

where

$$u_i | \beta, \sigma_u^2, \sigma_\epsilon^2, y_i \sim N \left\{ \tilde{u}_i; \left(z_i' z_i + \frac{\sigma_\epsilon^2}{\sigma_u^2} \right)^{-1} \sigma_\epsilon^2 \right\}$$

and

$$\tilde{u}_i = \left(z_i' z_i + \frac{\sigma_\epsilon^2}{\sigma_u^2} \right)^{-1} z_i' (y_i - X_i \beta).$$

The Gibbs sampler for $p(\beta, u, \sigma_\epsilon^2 | \sigma_u^2, y, M)$ can be described as follows (see also Kleinman and Ibrahim, 1998):

- (0) Select starting values for $u^{(0)}$ and $\sigma_\epsilon^{2(0)}$. Set $i = 0$.
- (1) Sample $\beta^{(i+1)}$ from $p(\beta | u^{(i)}, \sigma_\epsilon^{2(i)}, y)$ according to equation (2.5).
- (2) Sample $\sigma_\epsilon^{2(i+1)}$ from $p(\sigma_\epsilon^2 | \beta^{(i+1)}, u^{(i)}, y)$ according to equation (2.6).
- (3.1) Sample $u_1^{(i+1)}$ from $p\{u_1 | \beta^{(i+1)}, \sigma_\epsilon^{2(i+1)}, \sigma_u^2, u_{(1)}^{(i)}, y, M\}$ according to equation (2.9).
- $\vdots$
- (3.q) Sample $u_q^{(i+1)}$ from $p\{u_q | \beta^{(i+1)}, \sigma_\epsilon^{2(i+1)}, \sigma_u^2, u_{(q)}^{(i)}, y, M\}$ according to equation (2.9).
- (4) Set $i = i + 1$ and return to step (1).

For the Gibbs procedure described above, it is assumed that σ_u^2 and M are known. Typically the variance σ_u^2 in the base measure of the Dirichlet process in equation (2.3) is unknown and therefore a suitable prior distribution must be specified for it. Note that, once this has been accomplished the base measure is no longer marginally normal.

For convenience, suppose $p(\sigma_u^2) \propto \text{constant}$ to present lack of prior knowledge about σ_u^2 . After choosing random effects for each subject (sire), the subjects will be grouped into clusters in which the subjects have equal u_i 's. That is, after selecting a new u_i for each subject i in the sample, there will be some number k , $0 < k \leq q$, of unique values among the u_i 's. Denote these unique values by γ_ℓ , $\ell = 1, \dots, k$. Additionally let ℓ represent the set of subjects with common random effect γ_ℓ . Note that knowing the random effects is equivalent to knowing k , all of the γ 's and the cluster memberships ℓ . Then for the purposes of calculating the full conditionals of σ_u^2 , the γ_ℓ are k independent observations from $N(0, \sigma_u^2)$. Thus as in West et al. (1994) and Kleinman and Ibrahim (1998)

$$p(\sigma_u^2 | \gamma, y) = K_\gamma \left(\frac{1}{\sigma_u^2} \right)^{k/2} \exp \left\{ -\frac{1}{2\sigma_u^2} \gamma' \gamma \right\} \sigma_u^2 > 0 \quad (2.10)$$

an inverse gamma density, where

$$K_\gamma = \left\{ \frac{\gamma' \gamma}{2} \right\}^{\frac{1}{2}(k-2)} \frac{1}{\Gamma\left(\frac{k-2}{2}\right)}$$

and

$$\gamma = [\gamma_1 \ \gamma_2 \ \dots \ \gamma_k]'$$

The Gibbs sampling scheme illustrated above must therefore be modified by an insertion of a step to sample $\sigma_u^{2(i)}$ from the distribution (2.10).

Bush and MacEachern (1966) and Kleinman and Ibrahim (1998) also recommended one additional piece of the model as an aid to convergence for the Gibbs sampler. To speed mixing over the entire parameter space, they suggest moving around the γ 's after determining how the u_i 's are grouped.

The conditional posterior density of γ_ℓ is

$$p(\gamma_\ell | \beta, \sigma_\varepsilon^2, \sigma_u^2, y) \propto \phi(\gamma_\ell | 0, \sigma_u^2) \prod_{i \in \ell} p(y_i | \beta, \sigma_\varepsilon^2)$$

which implies that

$$\gamma_\ell | \beta, \sigma_\varepsilon^2, \sigma_u^2, y \sim N \left\{ \tilde{u}_\ell, \left(\sum_{i \in \ell} z'_i z_i + \frac{\sigma_\varepsilon^2}{\sigma_u^2} \right)^{-1} \sigma_\varepsilon^2 \right\} \quad (2.11)$$

where

$$\tilde{u}_\ell = \left(\sum_{i \in \ell} z'_i z_i + \frac{\sigma_\varepsilon^2}{\sigma_u^2} \right)^{-1} \left(\sum_{i \in \ell} z'_i (y_i - X_i \beta) \right).$$

The Gibbs sampling scheme is modified by inserting the following steps between (3.q) and (4).

(3a.1) Sample $\gamma_1^{(i+1)}$ from $p \left(\gamma_1 | \beta^{(i+1)}, \sigma_\varepsilon^{2(i+1)}, \sigma_u^{2(i)}, y \right)$ according to equation (2.11).

(3a.k) Sample $\gamma_k^{(i+1)}$ from $p \left(\gamma_k | \beta^{(i+1)}, \sigma_\varepsilon^{2(i+1)}, \sigma_u^{2(i)}, y \right)$ according to equation (2.11).

(4a) Sample σ_u^2 from $p \left(\sigma_u^2 | \gamma^{(i+1)}, y \right)$ according to equation (2.10).

3. Simulation of the precision parameter M (West, 1992)

The precision or total mass parameter M of the mixing Dirichlet process directly determines the prior distribution for k , the number of additional normal components in the mixture, and thus is a critical smoothing parameter for the model which in the previous sections have been assumed fixed or specified by the investigator (see Pretorius and Van der Merwe, 2000). In Antoniak (1974) it is shown that the prior distribution of k may be written as

$$p(k|M, q) = c_q(k) q^k M^k \frac{\Gamma(M)}{\Gamma(M+q)} \quad (k = 1, 2, \dots, q) \quad (3.1)$$

and $c_q(k) = p(k|M=1, q)$, not involving M . West (1992) mentioned that if required, the factors $c_q(k)$ can easily be computed using recurrence formulae for stirling numbers. It is also shown by him that the conditional

posterior distribution of M ,

$$p(M|k, \beta, \mathbf{u}, \sigma_e^2, \sigma_u^2, \mathbf{y}) = p(M|k) \propto p(M) p(k|M) \quad (3.2)$$

where $p(M)$ is the prior for M and the likelihood function is defined in (3.1). West (1992) assumed $M \sim G(a, b)$, a Gamma prior with shape $a > 0$ and scale $b > 0$ (which we may extend to include a reference prior (Uniform for $\log(M)$) by letting $a \rightarrow 0$ and $b \rightarrow 0$). In this note we will use the latter which means that

$$p(M) \propto M^{-1} \quad M > 0.$$

Equation (3.2) can be expressed as a mixture of two gamma posteriors, and the conditional distribution of the mixing parameter x given M and k is a simple beta. This can be illustrated as follows. For $M > 0$, the gamma functions in (3.1) can be written as

$$\frac{\Gamma(M)}{\Gamma(m+q)} \approx \frac{(M+q) B(M+1, q)}{M \Gamma(q)}$$

where $B(\cdot, \cdot)$ is the usual beta function. Then in (3.2) and for any $k = 1, 2, \dots, q$, the posterior

$$\begin{aligned} p(M|k) &\propto p(M) M^{k-1} (M+q) B(M+1, q) \\ &\propto M^{k-2} (M+q) \int_0^1 x^M (1-x)^{q-1} dx, \end{aligned} \quad (3.3)$$

using the definition of the beta function. From (3.3) it also follows that the joint posterior density of M and x is

$$p(M, x|k) \propto M^{k-2} (M+q) x^M (1-x)^{q-1} \quad (0 < M, 0 < x < 1) \quad (3.4)$$

and the conditional posteriors $p(M|x, k)$ and $p(x|M, k)$ can be determined as follows.

Firstly

$$\begin{aligned} p(M|x, k) &\propto M^{k-2} (M+q) e^{-M(-\log(x))} \\ &\propto M^{k-1} e^{-M(-\log(x))} + q M^{k-2} e^{-M(-\log(x))} \end{aligned}$$

for $M > 0$, which reduces easily to a mixture of two gamma densities, viz.

$$M|x, k \sim \pi_x G(k, -\log(x)) + (1 - \pi_x) G(k - 1, -\log(x)) \quad (3.5)$$

with weights π_x defined by $\frac{\pi_x}{1 - \pi_x} = \frac{k-1}{q(-\log(x))}$.

Secondly

$$p(x|M, k) \propto x^M (1-x)^{q-1} \quad 0 < x < 1 \quad (3.6)$$

so that $x|M, k \sim B(M+1, q)$, a beta distribution with mean $(M+1)/(M+q+1)$. Also $\log(x) = \log_e(x)$.

According to West (1992) it should now be clear how M can be sampled at each stage of the simulation: At each Gibbs iteration, the currently sampled values of k and M allow us to draw a new value of M by (i) first sampling an x value from the simple beta distribution (3.6), conditional on M and k fixed at their most recent values, then (ii) sampling the new M value from the mixture of gammas in (3.5) based on the same k and the x value just generated in (i).

The Gibbs sampling scheme described in section 2 can now be further modified by inserting the following steps between (4a) and (4).

(4b) Sample $x^{(i+1)}$ from $p(x|M^{(i)}, k^{(i)})$ according to equation (3.6).

(4c) Sample $M^{(i+1)}$ from $p(M|x^{(i+1)}, k^{(i)})$ according to equation (3.5).

On completion of the simulation, we will have a series of sampled values of k , M , x and all other parameters. Suppose that the Monte Carlo sample size is N , and denote the sampled values $k^{(i)}$, $x^{(i)}$, etc., for $i = 1, \dots, N$. Only the sampled values $k^{(i)}$ and $x^{(i)}$ are needed in estimating the posterior $P(M|y)$ via the usual Monte Carlo average of conditional posteriors, viz.

$$p(M|y) \simeq N^{-1} \sum_{i=1}^N p(M|x^{(i)}, k^{(i)})$$

where the summands are simply the conditional gamma mixtures in equation (3.5).

4. Results

Example: Animal breeding experiment to select for improved Helminth resistance in sheep (Duchateau et al., 1998).

The female sheep used in the experiment are from three different breeds, whereas the males are from two breeds. In each of the six crosses, there are at least 25 and at most 42 different sires. The weaning weight is measured for each of the lambs. The following fixed effects are included in the model: Age, sex and breed of female.

Results for different values of M are given and also where M is simulated from a mixture of Gamma distributions.

4.1 Variance components

The variance components obtained from REML, Traditional Bayes and the Dirichlet Process are reported in Table 1. The estimates are the posterior modes.

Table 1

	REML	Trad. Bayes	Dirichlet process			
			Simulated M	$M = 5$	$M = 100$	$M = 1000$
σ_e^2	4.8639	4.8885	4.8667	4.8915	4.8615	4.8743
σ_u^2 (Sire variance)	0.6802	0.7211	0.6308	0.6533	0.6588	0.7275

The estimated posterior densities of the sire variances for Trad. Bayes, Simulated M and $M = 5$ are given in Figure 1.

A commonly derived statistic, the heritability $h^2 = \frac{4\sigma_u^2}{\sigma_u^2 + \sigma_e^2}$, of the trait which is a function of the two variance components, is calculated and reported in Table 2. It describes the proportion of the total variation in the environment of this study attributable to inheritance from the lambs' parents. The estimates are the posterior modes.

Table 2

	REML	Trad. Bayes	Dirichlet process			
			Simulated M	$M = 5$	$M = 100$	$M = 1000$
h^2	0.49	0.5127	0.5097	0.4756	0.5179	0.5244

4.2 Fixed effects

Once estimates of the variance components have been obtained, they can be used to derive estimates for the fixed effects. These estimates are given in the REML column. the "Trad. Bayes" and "Dirichlet Process" estimates (posterior modes) were obtained using the Gibbs sampler. Only the results for the simulated M are reported (Dirichlet process) because unlike the densities of the random effects, the values of M have a minor effect on the posterior densities of the fixed effects.

Table 3

Effect		REML	Trad. Bayes	Dirichlet process
Intercept	(β_0)	5.003	4.344	4.327
Sex	(β_1)	-0.704	-0.705	-0.706
Breed 1	(β_2)	1.682	1.671	1.676
Breed 2	(β_3)	1.515	1.502	1.504
Breed 3	(β_4)	1.436	1.433	1.427
Breed 4	(β_5)	0.948	0.928	0.938
Breed 5	(β_6)	-0.406	-0.426	-0.408
Age	(β_7)	0.047	0.046	0.046

The REML estimates given in Table 3 are obtained by substituting the restricted maximum likelihood estimates of σ_u^2 and σ_e^2 in Henderson's mixed

model equations. For further information on these methods see for example Duchateau et al. (1998), Clarke (1998) and Henderson (1984).

From Table 3 (Dirichlet process) it is also clear that female lambs weigh on average 0.706 kg less at weaning than males. Furthermore, the weaning weight of a lamb increases by 0.046 kg per daily increase in age.

4.3 Random effects

Apart from estimating heritability, an investigator might be interested in the breeding value of specific sires in order to determine which sires should be retained for future selection. These breeding values for the first ten sires are listed in Table 4. The estimates are the posterior modes.

Table 4

Sire – ID	REML	Trad. Bayes	Dirichlet process		
			Simulated M	$M = 5$	$M = 1000$
1971	-0.10609	-0.1024	-0.109	-0.1307	-0.1063
1972	0.524072	0.5689	0.5081	0.3701	0.5265
1973	0.388752	0.4464	0.3934	0.2809	0.4164
1974	1.93394	1.9627	1.9026	1.7412	1.9434
1980	0.929863	0.9635	0.9193	0.8672	0.9327
1991	0.361078	0.3741	0.3438	0.1912	0.3200
1999	0.926551	0.9939	0.8936	0.9384	0.9314
4907	0.228914	0.2676	0.2293	0.0714	0.2512
4908	1.661416	1.6662	1.6490	1.6143	1.6740
4909	-0.7628	-0.7645	-0.7711	-0.7639	-0.7265

The predicted breeding value for sire with ID 1971 equals -0.109, meaning that offspring of this sire are predicted to have a weaning weight that is 0.109 kg less than the average weight of offspring of all sires of the same breed. Moreover from Figure 3(d) and Table 4 it is clear that the following of sire with

ID 1974, will have a predicted weaning weight of 1.9026 kg higher than the average offspring of all sires of the same breed. The 95% credibility interval on the other hand is 0.7862 – 3.1311 which is an indication that most (about all) of the offspring of this sire will have a weaning weight larger than the average.

The estimated posterior densities of the breeding values for sires 1971 and 1974 are given in Figures 2 and 3 and the difference in breeding values between these two sires are given in Figure 4.

Unlike the densities of the fixed effects, the values of M have a large effect on the posterior densities of the different breeding values. When $M = 5$ it is clear that there is some uncertainty about the exact location and height in the densities. We note from an examination of these plots that as M increases, the shapes of the densities tend to become more bell-shaped and symmetrical. This is also to be expected, as the random effects are directly affected by the relaxation of the normal assumptions (when M is small). If we let the value of M go down further, the number of clusters will definitely decrease and the posterior distributions of the random effects might tend to become somewhat skew or unstable.

The reason for displaying these densities is therefore mainly for illustrative and comparative purposes; i.e. to illustrate how the relaxation of the normal assumption will influence the distributions of the random effects. The correct posterior distributions of the random effects (breeding values) are the “simulated M ” densities and should be used in practice when necessary.

4.4 Precision parameter M

Figure 5 shows the observed histograms for k (the number of groups) for the different values of M while Table 5 below reports the different values (posterior modes) of k for $m = 5, 100, 1000$ and when M is simulated.

Table 5

	$M = 5$	$M = 100$	$M = 1000$	Simulated M
k	125	130	197	140

Figure 6 on the other hand shows the estimated posterior density of M when $k = 140$.

5. Discussion

It was mentioned by one of the referees that it is not clear at all what the contribution of the Bayesian methods to this data set or to the field brings because it seems to supply exactly the same answers as the “frequentist approach”.

In our opinion there are several shortcomings in the “classical” approach.

- (i) The properties of the predictors are hard to assess.
- (ii) When the values of the variance components are estimated from the data, their sampling errors are generally not taken into account in the subsequent analysis. Therefore, the variance of the prediction error will generally be underestimated.
- (iii) Depending upon the size and characteristics of the data, point estimators of variance components can be highly variable. For certain values of the variance component estimators, the predictors obtained by substituting these values in the “Best Linear Unbiased Predictor” are intuitively unappealing.
- (iv) Although the estimate for a variance component is always positive, the REML estimate’s asymptotic distribution can generate interval estimates that include negative values. This potentially embarrassing phenomenon is often overlooked in the discussions of the “frequentist approach”.

Some of the advantages of the Bayesian approach are:

- (i) The Bayesian practitioner does not need to commit himself to only a point estimate of the variance components. Credibility intervals can easily be obtained.

- (ii) Uncertainty about the true values of the variance components is formally incorporated into the analysis through the choice of an appropriate prior distribution.
- (iii) Given the data, the prior information about the unknown parameters and a well defined loss function, there exists an optimal Bayes predictor.
- (iv) All the available information about the random variable to be predicted is contained in the posterior distribution of the random variable. The practitioner can, therefore, base all of his inferences on this distribution.
- (v) The Bayesian approach is conceptually more appealing than the classical approach.
- (vi) Also highest posterior density regions are never empty, whereas confidence intervals for the ratio of two variance components can be empty. One can also report the whole of the posterior probability distribution, not just a single number, and report some measure of posterior precision. Finally, classical estimators generally have intractable sampling distributions and standard errors are hard to calculate.

The fact that the final comparisons (point and interval estimates) do not show much difference between the traditional Bayesian and non-Bayesian results is not surprising to us, since as showed by Harville (1974) (see also Searle, Casella and McCulloch, 1992) that when "flat" or uniform priors are assigned to the vector of fixed effects and the variance components, the modes of the marginal posterior distributions are very close to the REML estimates.

If, on the other hand, proper priors were assigned to the unknown parameters and if the sample size was quite small, the differences between the Bayesian and non-Bayesian results could have been quite substantial. The assignment of a proper prior to a specific parameter must, however, be justifiable from a practical point of view. In some animal breeding experiments, for example, it is known beforehand that $\sigma_u^2 < \frac{1}{3}\sigma_\epsilon^2$. This information can, if necessary, be used to formulate a proper prior on the interval

$[0; \frac{1}{3}]$ for the variance ratio $\tilde{\gamma} = \frac{\sigma_u^2}{\sigma_e^2}$. The Gibbs sampler can easily be adjusted to handle this case. The assignment of proper priors was, however, not the purpose of this note.

Both referees expressed some concern about the “simulated M ”. Since M is an unknown parameter, the first step in a Bayesian approach is to select a prior. If one has little prior information available, the “non-informative” default prior

$$p(M) \propto M^{-1} \quad M > 0$$

is appropriate. By using Gibbs sampling the unconditional posterior distribution of M can then be obtained. In our opinion this is the correct way of modelling M . If a point estimate of M is needed, the posterior mode, $M_0 = 220$ will be our choice and as an estimate of the number of clusters, $k_0 = 140$ will be used. The posterior distribution also indicates that there is a slight possibility that M can be as large as 300 or 400 and that the corresponding number of clusters or groups will then be 160 or more. The values for different M illustrated in the tables and displayed in Figures 1–5 are included mainly for illustrative and comparative purposes.

The fact that the REML, Trad. Bayes and Dirichlet process (simulated M) estimates given in Tables 1 to 4 are for all practical purposes the same, is also an indication that the normal prior assumption for the random effects is justifiable, which means that M and k (the number of clusters) will be large.

If, on the other hand, the number of clusters and simulated M are small for a specific animal breeding experiment, then the nonparametric Bayesian (Dirichlet process prior) estimates will differ more and will contribute more than the corresponding frequentist or traditional Bayesian results. Further non-symmetry in the case of the Dirichlet process prior can occur in more than one way. If M is small more sires will contribute to one u_i as

compared to another u_i . Also the assumption that the base measure G_0 for the random effects is normally distributed can be relaxed in many directions. By specifying the base measure as a scale mixture of normals; t -distributions, Double exponential, Exponential power family and Logistic distributions for example can then easily be obtained. Implementation of the Gibbs sampler will still be straightforward. More specific if $u_i | \sigma_u^2 \lambda_i \sim N(0, \lambda_i \sigma_u^2)$ and by placing a prior on λ_i , say $\nu \lambda_i^{-1} \sim \chi_\nu^2$, then $u_i | \sigma_u^2 \sim t_\nu(0, \sigma_u^2)$, a t -distribution with mean 0, variance proportional to σ_u^2 and ν degrees of freedom. Skew distributions, for example Extreme value distributions, can also be considered as base measure.

6. Conclusion

It was mentioned in the introductory paragraph that the parametric form of the distribution of random effects can be a severe constraint. There is also a general feeling among many statisticians that it is desirable in many contexts to make fewer assumptions about the underlying populations from which the data are obtained than are required for a parametric analysis. In this manuscript we have applied a general technique for Bayesian nonparametrics to an important class of models, the mixed linear model in the case of animal breeding experiments. The important contribution of this paper revolves around the nonparametric prior distribution, the Dirichlet process prior, for the random effects and to model and interpret the estimated effects correctly. It is also clear from the example that the value of the precision parameter M largely influences the posterior distribution of the random effects, the sire variance and the heritability coefficient. The value of M will further determine whether the estimates from the Dirichlet process behave like the standard Bayes estimate or like a nonparametric likelihood estimator. Further research and simulation

experiments should be done to determine if the Dirichlet process prior can be used for solving the “undeclared preferential treatment of animals” problem.

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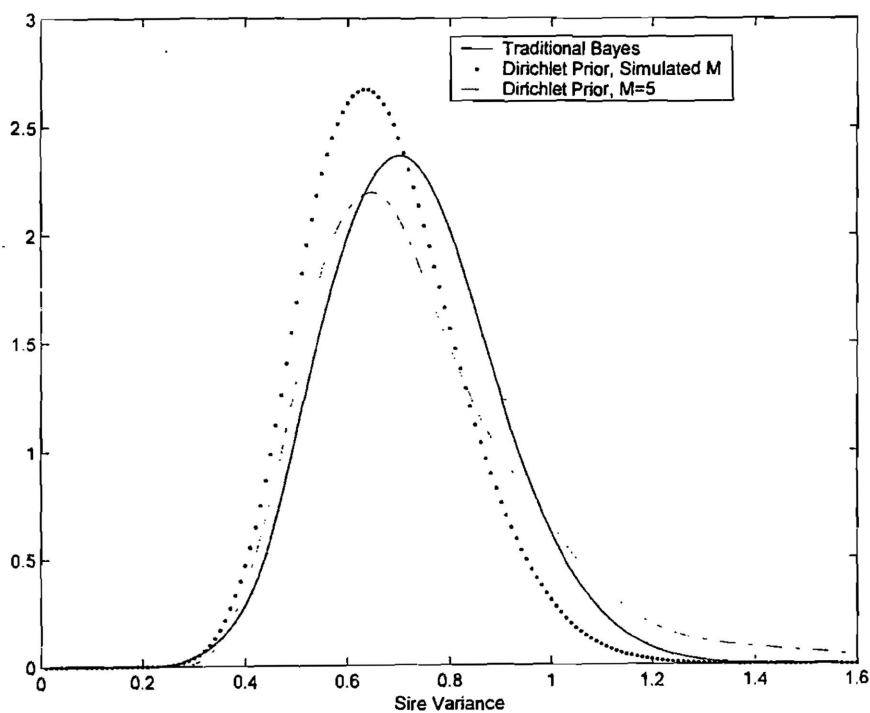


Figure 1. Comparison of estimates for the sire variance.

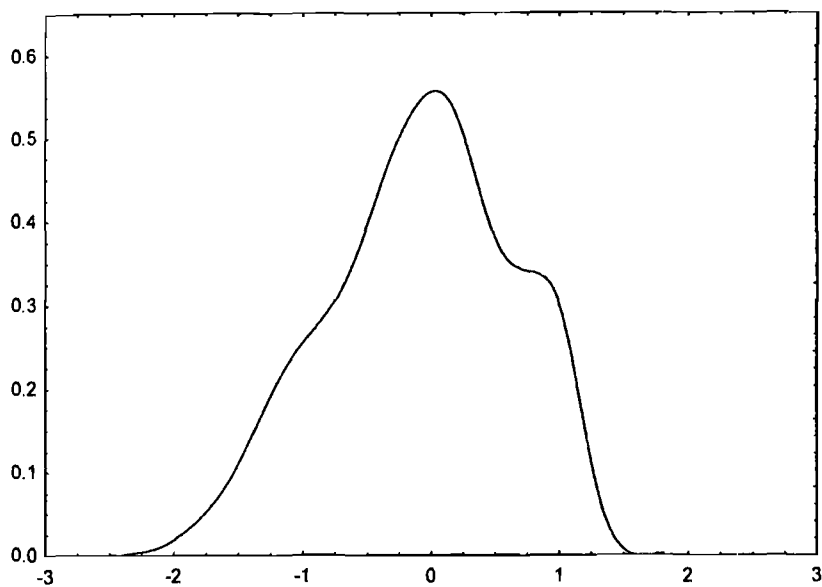


Figure 2a. The estimated marginal posterior density of the breeding value for Sire ID1971 when $M = 5$ in the Dirichlet process

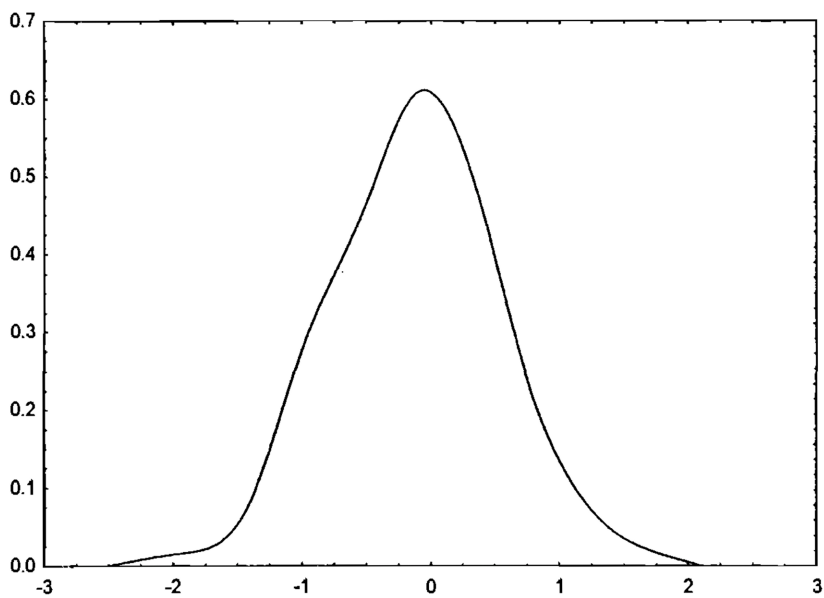


Figure 2b. The estimated marginal posterior density of the breeding value for Sire ID1971 when $M = 50$ in the Dirichlet process

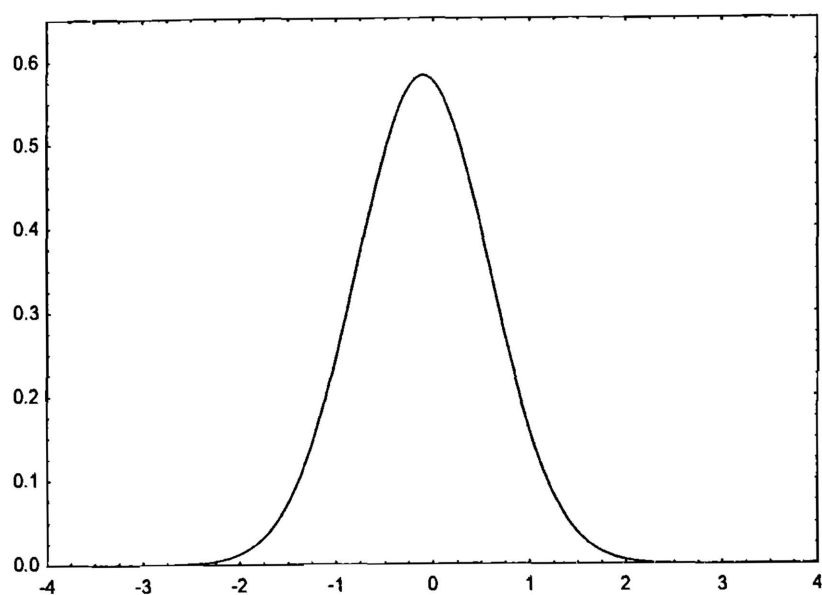


Figure 2c. The estimated marginal posterior density of the breeding value for Sire ID1971 when $M = 100$ in the Dirichlet process

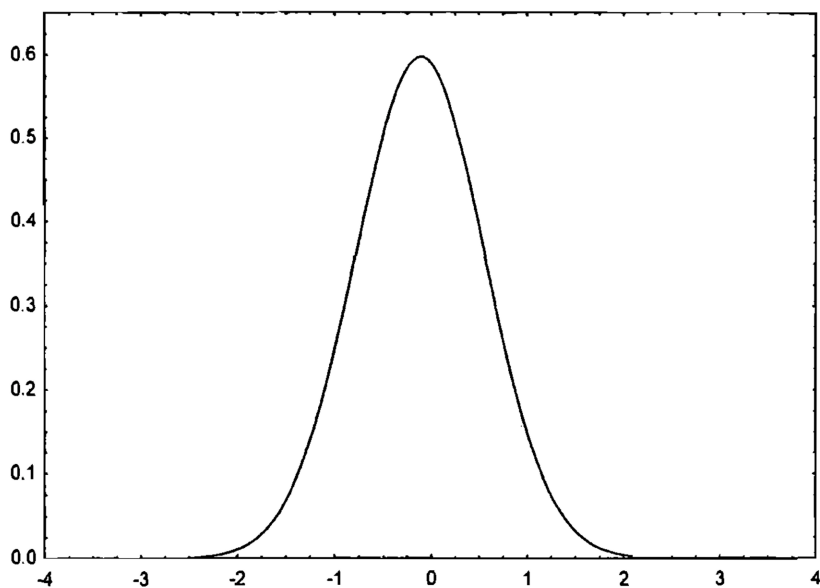


Figure 2d. The estimated marginal posterior density of the breeding value for Sire ID1971 when M is simulated in the Dirichlet process

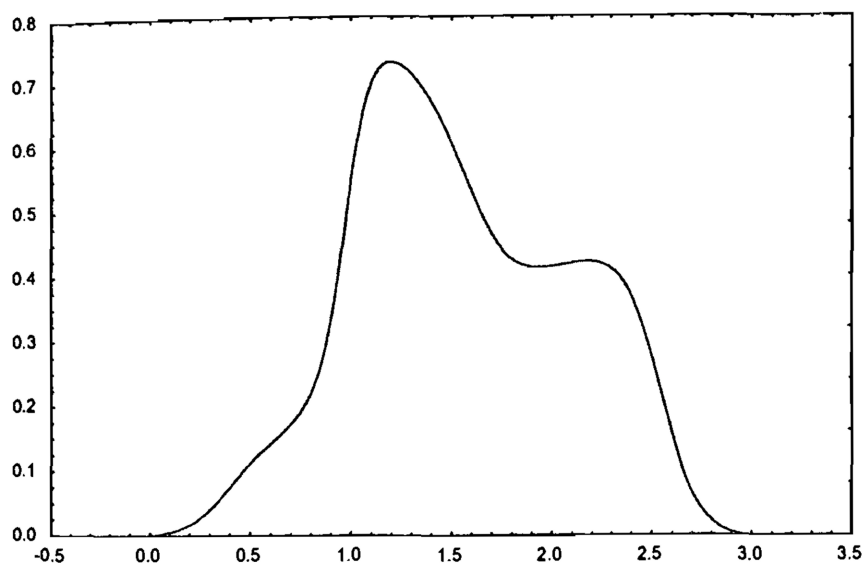


Figure 3a. The estimated marginal posterior density of the breeding value for Sire ID1974 when $M = 5$ in the Dirichlet process

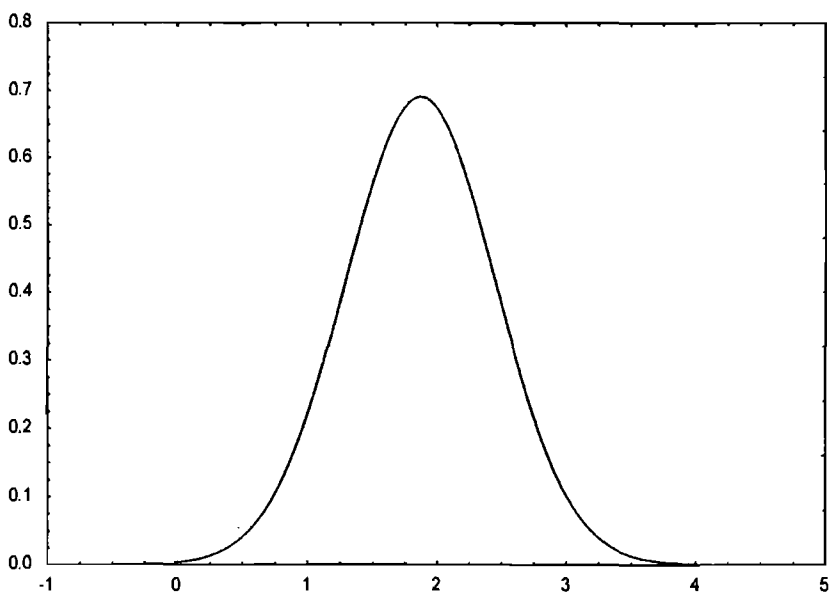


Figure 3b. The estimated marginal posterior density of the breeding value for Sire ID1974 when $M = 50$ in the Dirichlet process

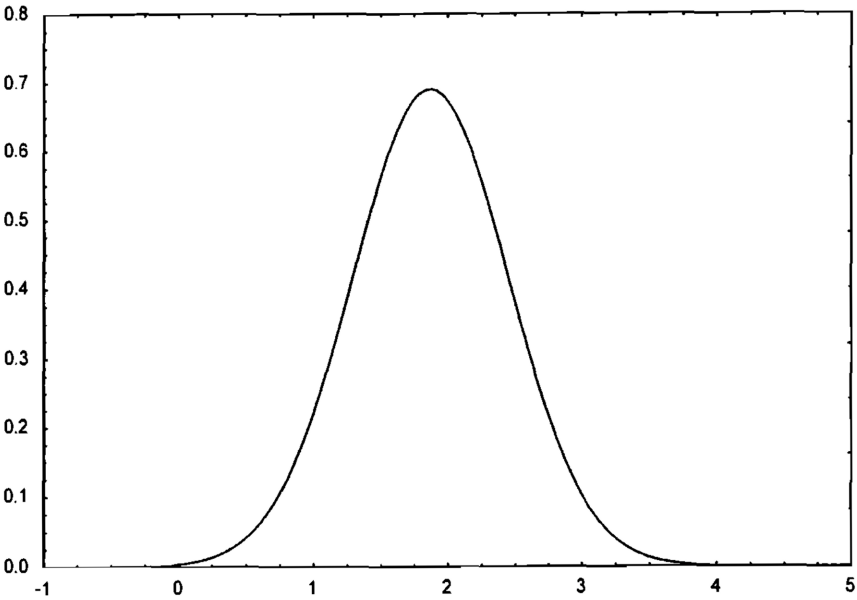


Figure 3c. The estimated marginal posterior density of the breeding value for Sire ID1974 when $M = 100$ in the Dirichlet process

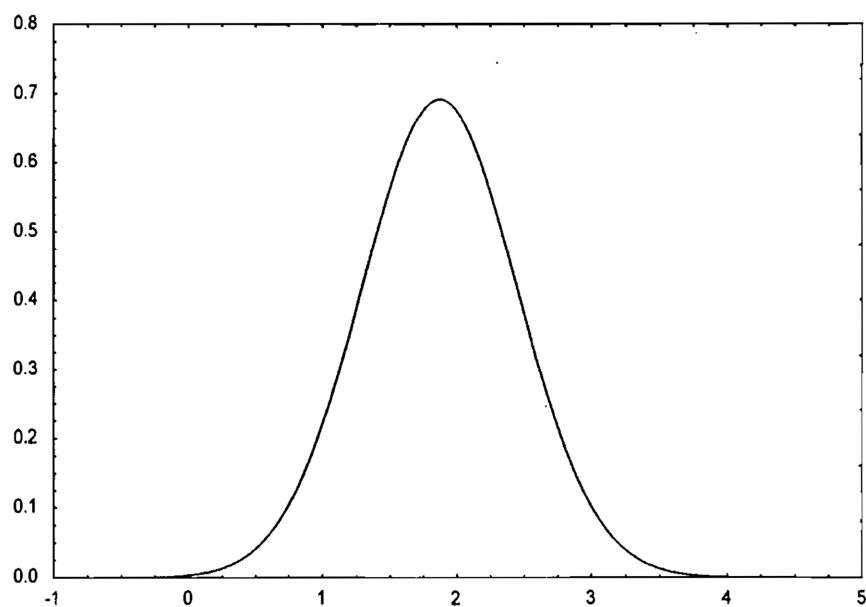


Figure 3d. The estimated marginal posterior density of the breeding value for Sire ID 1974 when $M = 5$ is simulated in the Dirichlet process

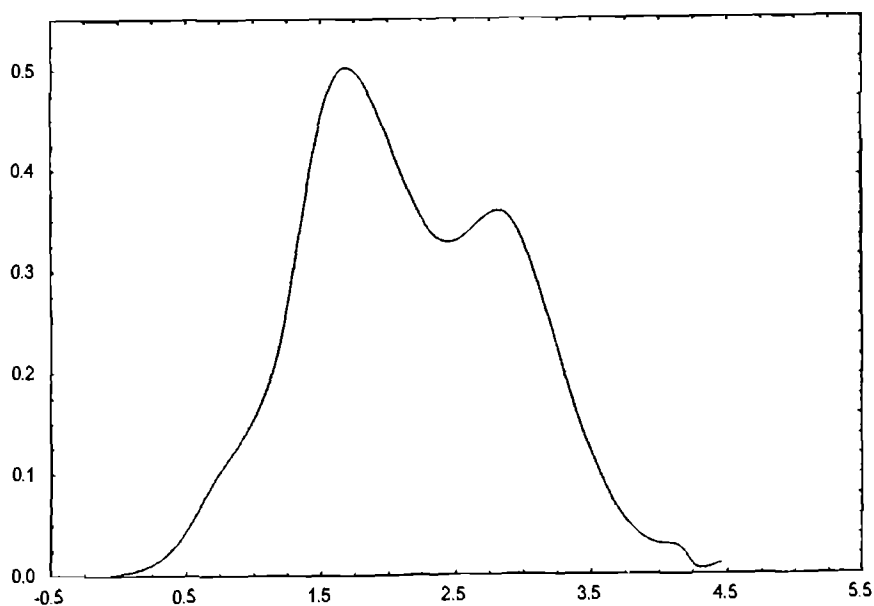


Figure 4a. Estimated marginal posterior density of the difference between breeding values for Sire ID1974 and Sire ID1971 when $M = 5$ in the Dirichlet process

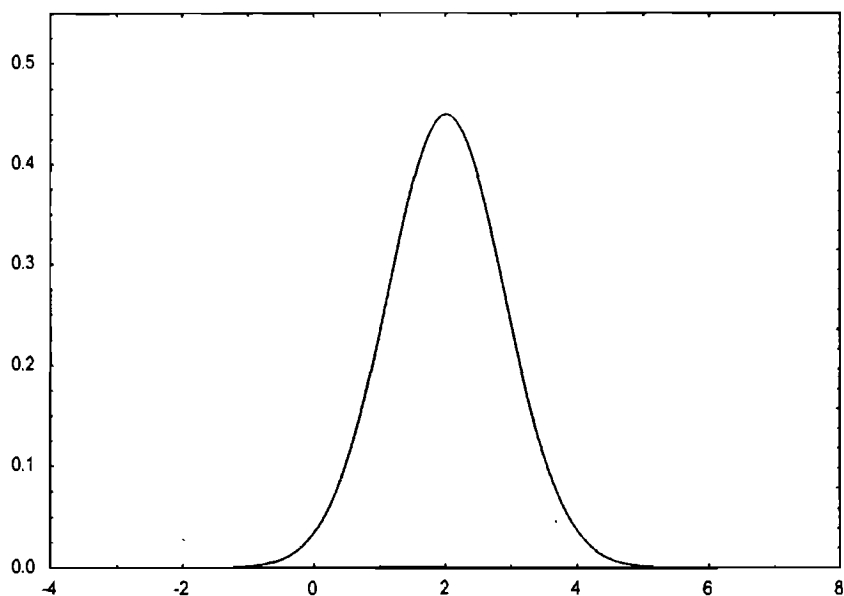


Figure 4b. Estimated marginal posterior density of the difference between breeding values for Sire ID1974 and Sire ID1971 when M is simulated in the Dirichlet process

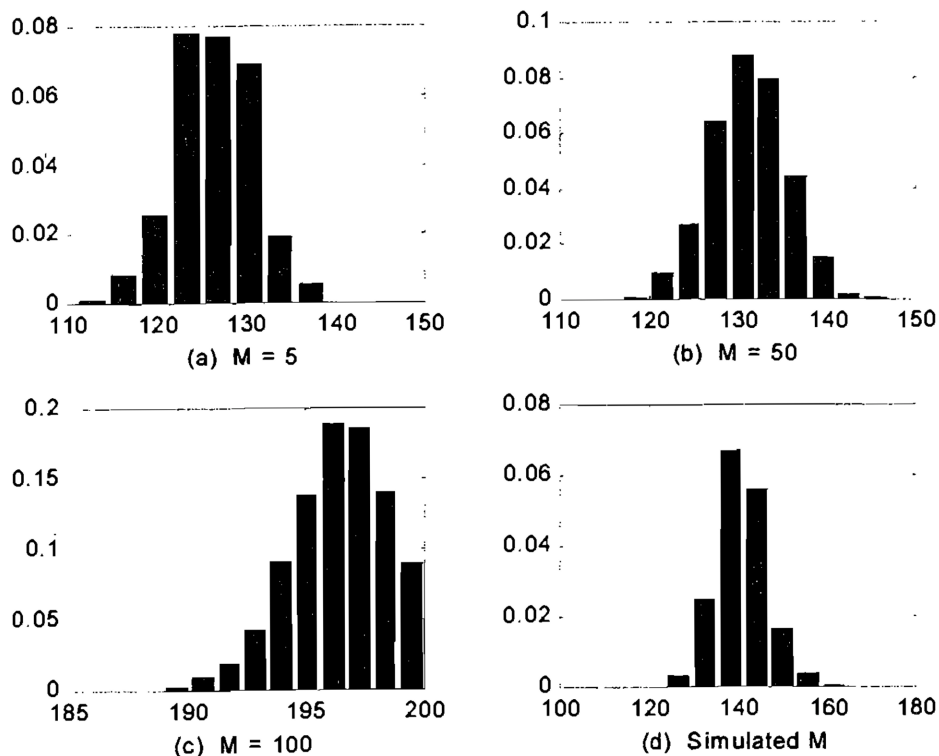


Figure 5. Histograms for k , the number of groups

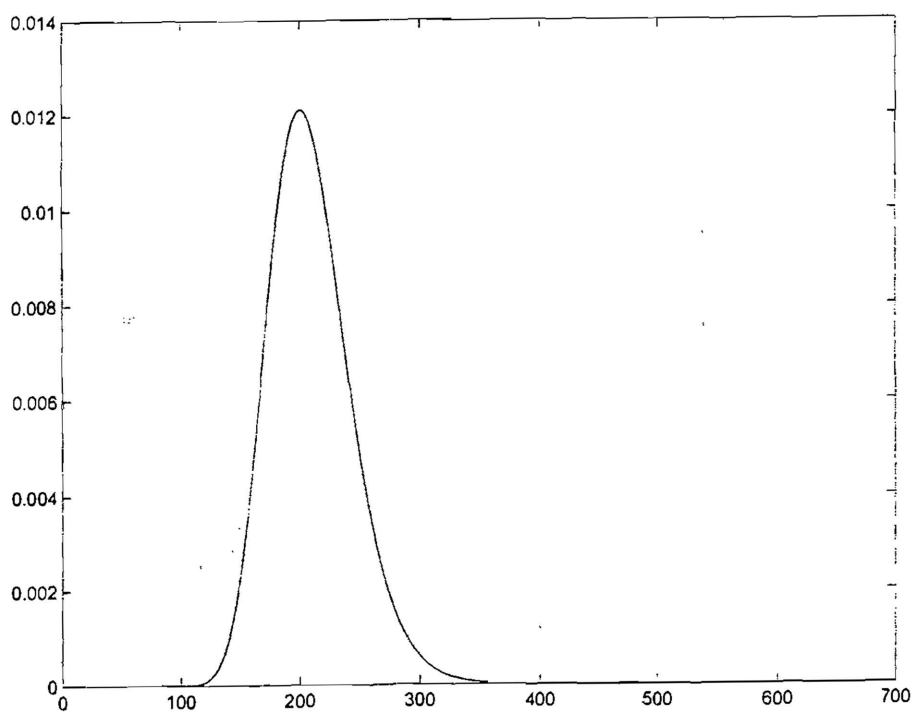


Figure 6. Estimated density of M for $k = 140$. Posterior mode = 220.